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Q#1 List down 20 real life sentences and apply unary, binary & high order constraints.

Q#2 implement Hill climbing

Q#3 20 important Qstns from code.

Q#4 Create graph of Pakistan such as meter, add, isld, Lhr, G.B etc apply simple backtracking technique on graph, also list variable domains, constra

UNARY:

1. A person must be over 18 to vote.
2. The shop opens after 9 AM
3. A student can take only 5 courses
4. The password must contain only 5 digits
5) The room temperature must stay below 20
6) A car cannot exceed 100 mph.
7) Employee must wear uniforms.

Binary:

8. Ali must sit nex to Ahmed.
9) Two meetings cannot occur at the same time.
10) if it rains, the match is cancelled.
11) A printer connects to one computer.
12) The teacher checks each student

assignment.
13) one kei
14) Driver m
registred
Hi
15) Three f
same
16) Math, P
cannot be
17) A projec
and Lec
18) Father
sit in
19) Breakfast
must Star
20) Teamas
same

Q#3 20
1: What
As ear
2: what
cross
7 What
Present

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real life sentence
binary g in high

Q Climbing

astus from code.

of Pakistan such
Ld, Lhr, G.P etc
technique on
domains, tourism

is to vote.
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d 100/mph.
wear uniforms.

ex to Ahmed.
cannot occur at

n is candles.
computer.
h s!

Mo Tu We Th Fr Sa Su

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assignment.
13) One key opens one specific lock
14) Driver ~~must~~ and vehicle must be registered together

High Order:

15) Three friends must travel in the same car.
16) Math, physics and english exam cannot be on one day.
17) A project requires a manager, develo and design togather.
18) Father, mother and child, must sit in the same row.
19) Breakfast, launch and dinner must stay with in the calorie limit
20) Teams A, B, C cannot use the same ground simultaneously.

Q#3 20 important Q stus from code

1: What is hill climbing:
A search optimazation algorithm.

2: What is start point:
current = 0

3 What does current represent?
Present state.

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4) What is next state?
Neighbours state.

5) How is next state generated?
current + 1

6) what loop is used?
while true

7) what is the goal funtion here
maximize x^2.

8) What condition implies state?
next_state i = 5

9) what happens if condition is true?
move to better state.

10) what happens if no better state exists?
Loop stops

11) which statement stops loop
break

12) what it finall returns
best state.

Mo Tu We Th Fr Sa Su

13) what

14) wh
L

15) Does
all p

16) what

17) wh
moves

18) when
this
curr

19) is
com

20) wh
Bes

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13) what is the final result
5

14) which search technique is used
Local search

15) Does hill climbing stores all paths?
No.

16) what type of algorithm is it?
Greedy algorithm

17) why it is called hill climb
moves toward better value

18) what is the neighbour in this example?
current +1

19) is this algorithm simple or complex?
simple.

20) what is printed at end?
Best state: 5

Q = 4

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Pakistan city graph:

Abbotabad (Red)
|
Islamabad (green)
/ \
Lahore (Red) — Multan (blue)
|
Karachi (Green)

Variables:
A = Abotabad
I = Islamabad
L = Lahore
M = Multan
K = Karachi

Domains
(Red, green, blue)

Constraints
1 Abbotabad ≠ Islamabad
2 Islamabad ≠ Lahore
3 Islamabad ≠ multan
4 Lahore ≠ multan
5 Lahore ≠ Karachi